Huawei Access Node WFD-ACC-09 Routing Protocol Flap SOP

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5365

Description

This Standard Operating Procedure (SOP) is designed to address the occurrence of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Huawei Access Node WFD-ACC-09 at the Whitefield POP. The Routing Protocol Flap failure domain is characterized by rapid oscillations in the routing protocol adjacency states, resulting in a cascade of network instability and subsequent service disruptions. This phenomenon is primarily attributed to the dynamic nature of routing protocols, which relies on the exchange of routing information between neighboring nodes. In the context of Huawei's Access Node WFD-ACC-09, this failure domain is likely to occur due to misconfigurations, hardware malfunctions, or software bugs that compromise the stability of the routing protocol. The BGP route churn, OSPF adjacency loss, and ISIS LSP flooding are all symptoms of a deeper issue, often related to the inconsistent or unreliable exchange of routing information between nodes. This SOP aims to provide a structured approach to diagnosing and resolving the Routing Protocol Flap failure domain on the Huawei Access Node WFD-ACC-09.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible via remote hands
- Check the status of the redundant links and ensure they are operational
- Review the system logs for any recent errors or anomalies

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor	High
Hardware malfunction in the routing engine	Medium
Software bug in the routing protocol stack	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp neighbor, Expected Output: show neighbor state, address, and timer information, Description: This command displays the BGP neighbor information, including the state, address, and timer information. It helps to identify any misconfigured BGP neighbors or issues with the neighbor relationships.

Command: display ospf peer, Expected Output: show OSPF peer state, IP address, and timer information, Description: This command displays the OSPF peer information, including the state, IP address, and timer information. It helps to identify any issues with the OSPF adjacency
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or neighbor relationships.

Command: display isis lsp, Expected Output: show ISIS LSP state, IP address, and sequence number information, Description: This command displays the ISIS LSP information, including the state, IP address, and sequence number information. It helps to identify any issues with the ISIS LSP flooding or neighbor relationships.
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Resolution Steps

Step 1: Step: Clear BGP neighbor, Command: clear bgp neighbor , Description: This command clears the BGP neighbor relationship and resets the BGP session. It helps to resolve issues related to misconfigured BGP neighbors or issues with the neighbor relationships.

Step 2: Step: Restart the routing engine, Command: reboot, Description: This command restarts the routing engine and resets the system. It helps to resolve issues related to hardware malfunctions or software bugs in the routing protocol stack.

Step 3: Step: Upgrade the routing protocol software, Command: system software upgrade, Description: This command upgrades the routing protocol software to the latest version. It helps to resolve issues related to software bugs in the routing protocol stack.

Post-Check Verification

```
Command: display bgp neighbor, Expected Output: show neighbor state, address, and timer information with no errors or anomalies, Description: This command verifies that the BGP neighbor information is correct and there are no errors or anomalies.

Command: display ospf peer, Expected Output: show OSPF peer state, IP address, and timer information with no errors or anomalies, Description: This command verifies that the OSPF peer information is correct and there are no errors or anomalies.

Command: display isis lsp, Expected Output: show ISIS LSP state, IP address, and sequence number information with no errors or anomalies, Description: This command verifies that the ISIS LSP information is correct and there are no errors or anomalies.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: undo , Description: This command restores the original configuration and reverts any changes made during the troubleshooting process.

- Step: Reset the system to its previous state, Command: system reset, Description: This command resets the system to its previous state and reverts any changes made during the troubleshooting process.